



IPsec VPN

Penetration Testing with BackTrack Tools

This article outlines the value of penetration-testing VPN gateways for known vulnerabilities and also shows you how to prevent a breach into the internal network.

IPsec is the most commonly used technology for both gateway-to-gateway (LAN-to-LAN) and host to gateway (remote access) enterprise VPN solutions. It offers complete data protection for tunnelled traffic, with confidentiality, integrity, data origin authentication and anti-replay services. IPsec uses a lot of components to achieve high-level security. The major protocols that IPsec uses are:

- **ESP (Encapsulation Security Payload):** ESP can provide data confidentiality and integrity, but cannot protect the IP header. The IP protocol number of ESP is 50.
- **AH (Authentication Header):** AH can provide the integrity service to the data packet, but cannot offer confidentiality to data packets like ESP. The IP protocol number of AH is 51.
- **IKE (Internet Key Exchange):** IKE provides support for the negotiation of parameters between endpoints or VPN peers and thus establishes, maintains and terminates security associations (SA). The SA termination can be based on time (seconds) or transfer

(kilobytes) rate. Actually, IKE is a type of ISAKMP (Internet Security Association Key Management Protocol) implementation, which is a framework for authentication and key exchange.

IKE establishes the security association (SA) between two endpoints through a three-phase process.

IKE Phase 1

IKE Phase 1 sets up a secure channel between two IPsec endpoints by the negotiation of parameters like the encryption algorithm, integrity algorithm, authentication type, key distribution mechanism, life time, etc. IKE Phase 1 can either use the main mode or aggressive mode to establish the bi-directional security association. Main mode negotiates SA through three pairs of messages, while aggressive mode offers faster operations through the exchange of three messages.

IKE Phase 2

IKE Phase 2 is used for data protection. The VPN